In [517]:

```
install.packages("readr")
library(readr)

data=read_csv("World_population(1951-2020).csv")
head(data)
str(data)
```

```
The downloaded binary packages are in
 /var/folders/01/_bgw0fxs5rv8ysw_gs46q0v40000gn/T//RtmpldRZXz/downloaded_packages
```

```
Rows: 70 Columns: 9
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (2): Yearly Change, Urban Pop %
dbl (4): Year, Density (P/Km²), Median Age, Fertility Rate
num (3): World Population, Net Change, Urban Pop

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

A tibble: 6 × 9

Year	World Population	Yearly Change	Net Change	Density (P/Km²)	Urban Pop	Urban Pop %	Median Age	Fertility Rate
<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
2020	7794798739	1.05%	81330639	52	4378993944	56%	30.9	2.47
2019	7713468100	1.08%	82377060	52	4299438618	56%	29.8	2.51
2018	7631091040	1.10%	83232115	51	4219817318	55%	29.8	2.51
2017	7547858925	1.12%	83836876	51	4140188594	55%	29.8	2.51
2016	7464022049	1.14%	84224910	50	4060652683	54%	29.8	2.51
2015	7379797139	1.16%	84506374	50	3981497663	54%	30.0	2.52

```
spc_tbl_ [70 × 9] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ Year            : num [1:70] 2020 2019 2018 2017 2016 ...
 $ World Population: num [1:70] 7.79e+09 7.71e+09 7.63e+09 7.55e+09 7.46e+09 ...
 $ Yearly Change   : chr [1:70] "1.05%" "1.08%" "1.10%" "1.12%" ...
 $ Net Change      : num [1:70] 81330639 82377060 83232115 83836876 84224910 ...
 $ Density (P/Km²) : num [1:70] 52 52 51 51 50 50 49 48 48 47 ...
 $ Urban Pop       : num [1:70] 4.38e+09 4.30e+09 4.22e+09 4.14e+09 4.06e+09 ...
 $ Urban Pop %     : chr [1:70] "56%" "56%" "55%" "55%" ...
 $ Median Age      : num [1:70] 30.9 29.8 29.8 29.8 29.8 30 NA NA NA NA ...
 $ Fertility Rate  : num [1:70] 2.47 2.51 2.51 2.51 2.51 2.52 NA NA NA NA ...
 - attr(*, "spec")=
  .. cols(
  ..   Year = col_double(),
  ..   `World Population` = col_number(),
  ..   `Yearly Change` = col_character(),
  ..   `Net Change` = col_number(),
  ..   `Density (P/Km²)` = col_double(),
  ..   `Urban Pop` = col_number(),
  ..   `Urban Pop %` = col_character(),
  ..   `Median Age` = col_double(),
  ..   `Fertility Rate` = col_double()
  .. )
 - attr(*, "problems")=<externalptr>
```

In [518]:

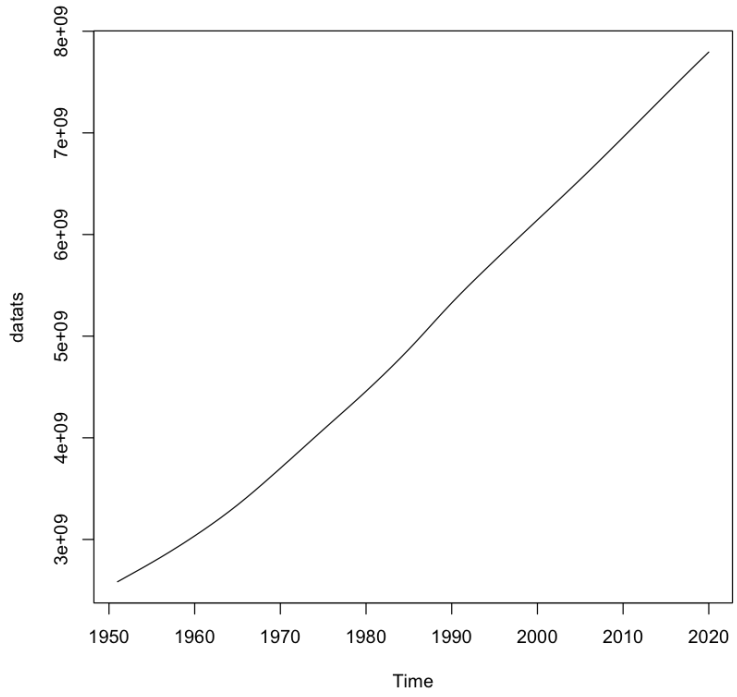
```
install.packages("dplyr")
library(dplyr)
#data['World Population']=log(data["World Population"])
# Arrange the data by the "Year" column in ascending order
data_sorted <- arrange(data, Year)
```

```
# Extract the "World Population" column and create a time series object
datats <- ts(data_sorted$`World Population`,start=c(1951))
```

```
The downloaded binary packages are in
 /var/folders/01/_bgw0fxs5rv8ysw_gs46q0v40000gn/T//RtmpldRZXz/downloaded_packages
```

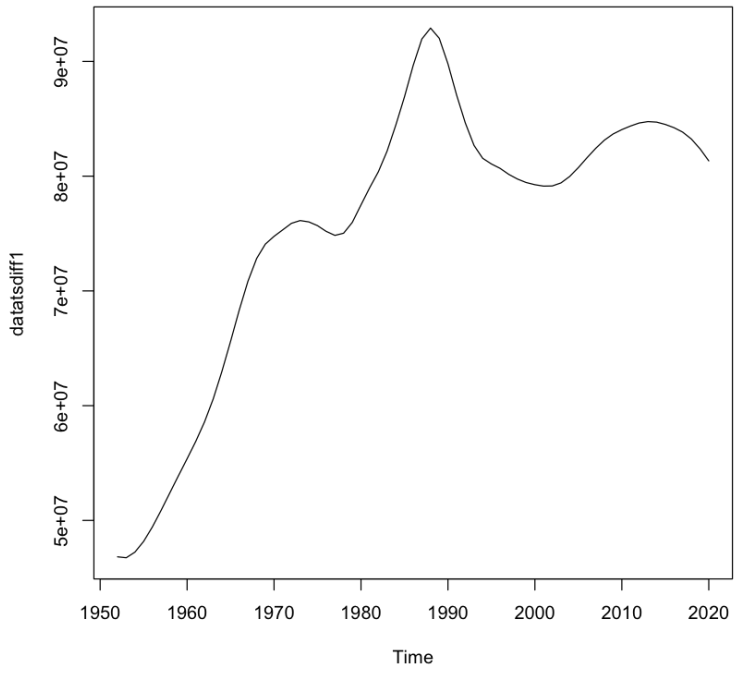
In [519]:

```
# Plot the time series
plot.ts(datats)
```



In [520]:

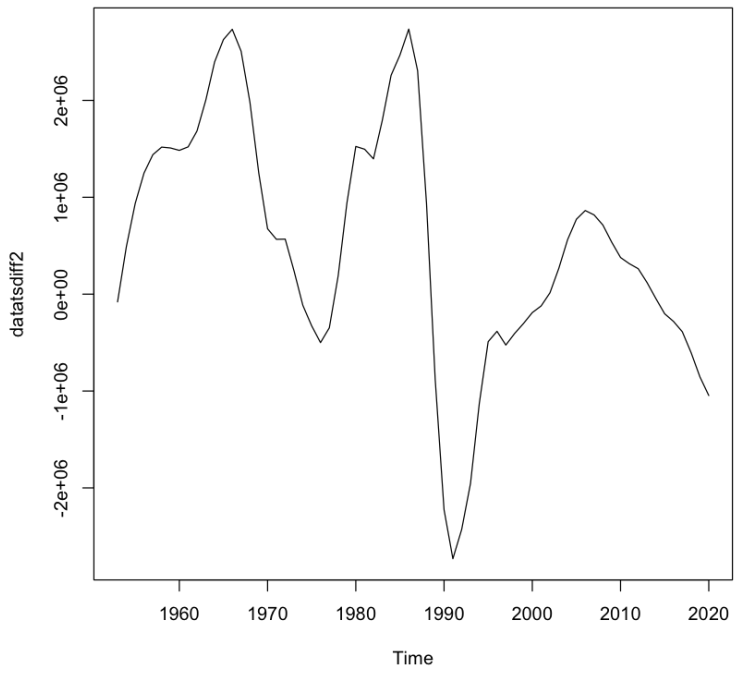
```
datatsdiff1=diff(datats,differences=1)
plot.ts(datatsdiff1)
```



In [521]:

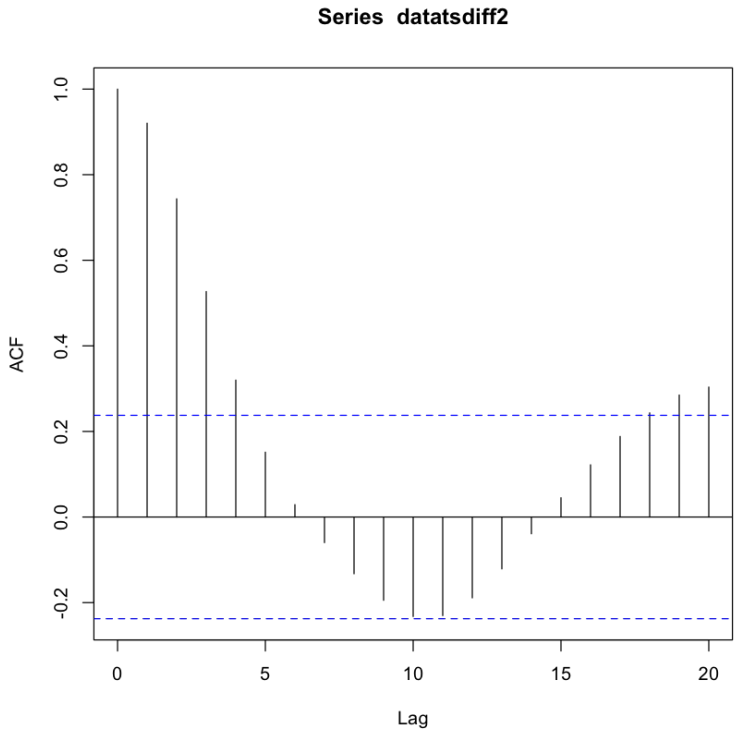
```
datatsdiff2=diff(datats,differences=2)
```

```
plot.ts(datatsdiff2)
```



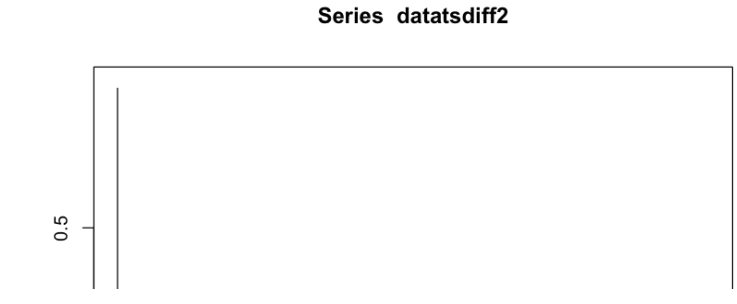
In [522]:

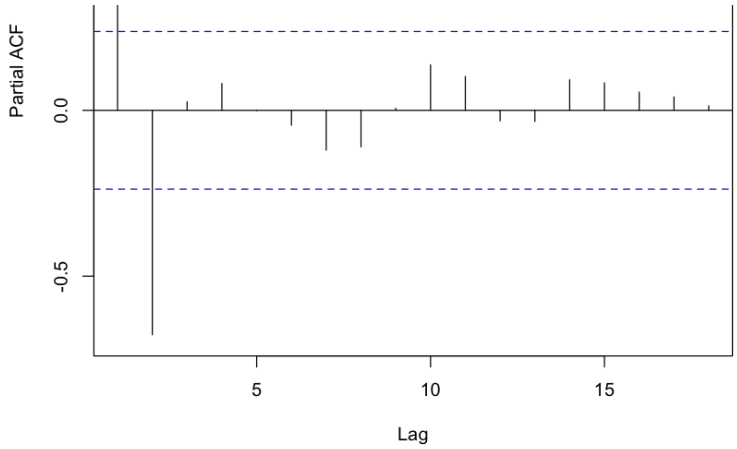
```
acf(datatsdiff2,lag.max=20)
```



In [523]:

```
pacf(datatsdiff2)
```





In [524]:

```
arima1=arima(datats,order=c(2,2,0))
arima1
```

```
Call:
arima(x = datats, order = c(2, 2, 0))

Coefficients:
         ar1      ar2
      1.7268  -0.8314
s.e.  0.0623   0.0620

sigma^2 estimated as 5.77e+10:  log likelihood = -941.23,  aic = 1888.46
```

In [525]:

```
library(lmtest)
```

In [526]:

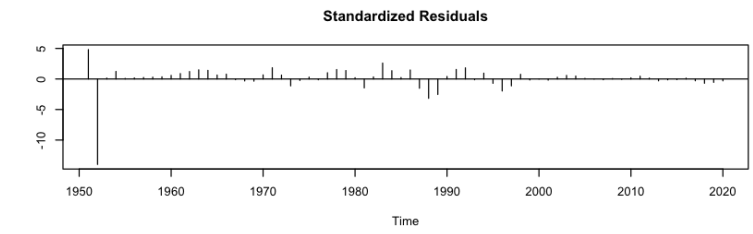
```
coeftest(arima1)
```

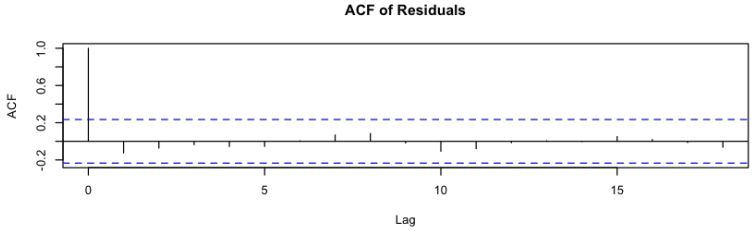
```
z test of coefficients:

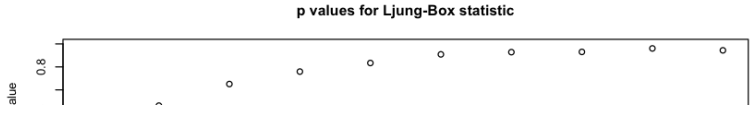
     Estimate Std. Error z value  Pr(>|z|)
ar1  1.726805   0.062292  27.721 < 2.2e-16 ***
ar2 -0.831435   0.062044 -13.401 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

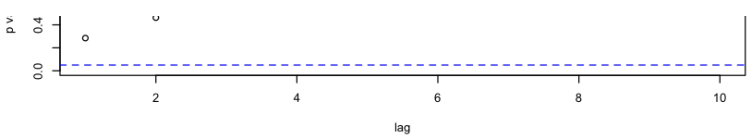
In [527]:

```
tsdiag(arima1)
```



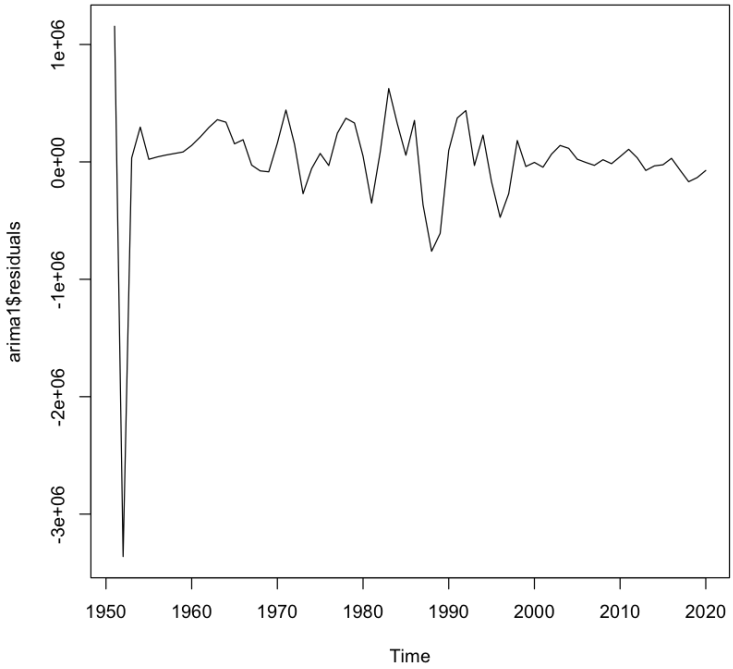






In [528]:

```
plot.ts(arima1$residuals)
```



In [552]:

```
install.packages("bestNormalize")
library(bestNormalize)
```

```
also installing the dependencies ‘RcppParallel’, ‘lamW’, ‘rngtools’, ‘lobstr’, ‘LambertW’
, ‘nortest’, ‘doParallel’, ‘doRNG’, ‘butcher’
```

```
The downloaded binary packages are in
 /var/folders/01/_bgw0fxs5rv8ysw_gs46q0v40000gn/T//RtmpldRZXz/downloaded_packages
```

```
Attaching package: ‘bestNormalize’


The following object is masked from ‘package:MASS’:

    boxcox
```

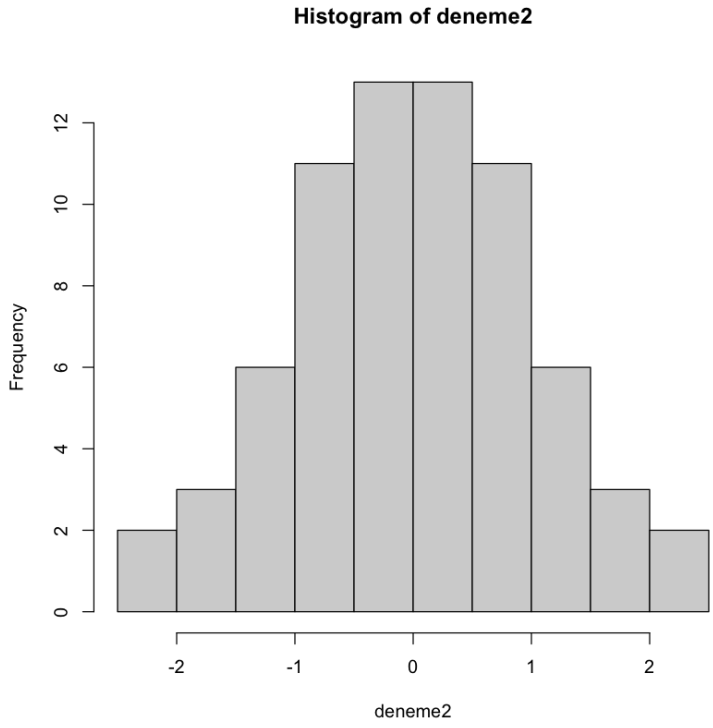
In [561]:

```
deneme=bestNormalize(arima1$residuals,allow_lambert_s=TRUE)
deneme
deneme2=predict(deneme)
hist(deneme2)
```

```
Best Normalizing transformation with 70 Observations
 Estimated Normality Statistics (Pearson P / df, lower => more normal):
 - arcsinh(x): 4.9429
 - Center+scale: 2.1857
 - Double Reversed Log_b(x+a): 1.8381
 - Log_b(x+a): 1.7905
 - orderNorm (ORQ): 1.7571
 - sqrt(x + a): 1.8333
```

```
 - Yeo-Johnson: 2.3
Estimation method: Out-of-sample via CV with 10 folds and 5 repeats

Based off these, bestNormalize chose:
orderNorm Transformation with 70 nonmissing obs and no ties
 - Original quantiles:
         0%         25%         50%         75%        100%
-3361712.54   -43721.55    43810.03   176687.15  1155611.88
```



In [567]:

```
datats2=predict(deneme,newdata=datats,inverse=TRUE)
plot.ts(datats2)
```

```
Warning message in inv_orderNorm_trans(object, newdata[!na_idx], warn):
"Transformations requested outside observed domain; logit approx. on ranks applied"
Warning message in min(x):
"min için eksik olmayan argüman yok; Inf döndürülüyor"
Warning message in max(x):
"max için eksik olmayan argüman yok; -Inf döndürülüyor"
```

```
Error in plot.window(xlim, ylim, log, ...): sonlu 'ylim' değerleri gerekli
Traceback:

1. plot.ts(datats2)
2. plotts(x = x, y = y, plot.type = plot.type, xy.labels = xy.labels,
 .     xy.lines = xy.lines, panel = panel, nc = nc, xlabel = xlabel,
 .     ylabel = ylabel, axes = axes, ...)
3. plot.window(xlim, ylim, log, ...)
```

In [531]:

```
Box.test(arima1$residuals,lag=6)
```

```
 Box-Pierce test

data:  arima1$residuals
X-squared = 1.9926, df = 6, p-value = 0.9204
```

In [532]:

```
Box.test(arima1$residuals,lag=12)
```

```
 Box-Pierce test

data:  arima1$residuals
X-squared = 4.1144, df = 12, p-value = 0.9813
```

In [533]:

```
Box.test(arima1$residuals,lag=18)
```

```
 Box-Pierce test

data:  arima1$residuals
X-squared = 4.6243, df = 18, p-value = 0.9993
```

In [534]:

```
Box.test(arima1$residuals,lag=24)
```

```
 Box-Pierce test

data:  arima1$residuals
X-squared = 5.9598, df = 24, p-value = 0.9999
```

In [535]:

```
arima2=arima(datats,order=c(0,2,5))
arima2
```

```
Call:
arima(x = datats, order = c(0, 2, 5))

Coefficients:
         ma1     ma2     ma3     ma4     ma5
      2.8090  4.0141  3.5240  1.9239  0.4384
s.e.  0.1239  0.3515  0.4647  0.3211  0.1157

sigma^2 estimated as 2.89e+10:  log likelihood = -922.27,  aic = 1856.54
```

In [536]:

```
coeftest(arima2)
```

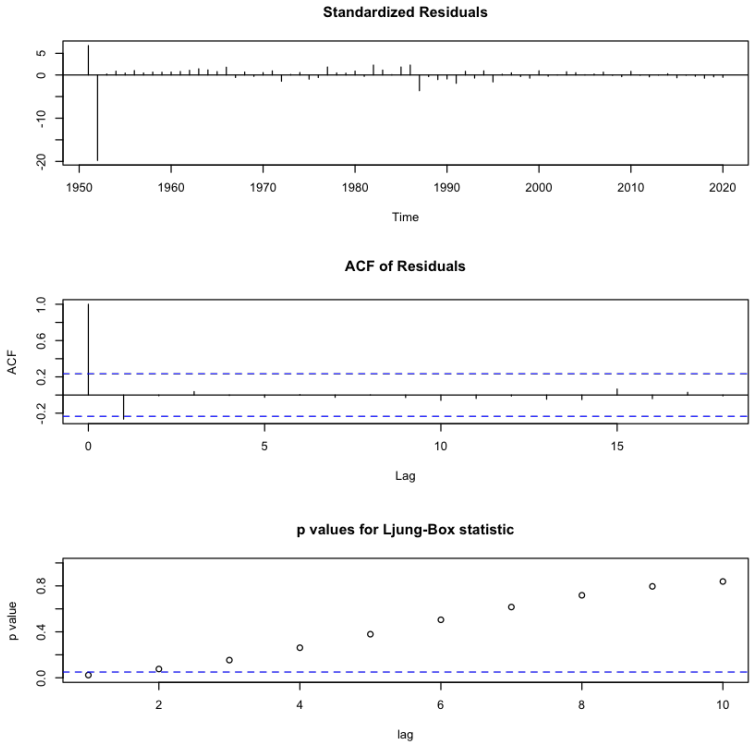
```
z test of coefficients:

    Estimate Std. Error z value  Pr(>|z|)
ma1  2.80896    0.12388 22.6756 < 2.2e-16 ***
ma2  4.01413    0.35151 11.4197 < 2.2e-16 ***
ma3  3.52402    0.46473  7.5829 3.378e-14 ***
ma4  1.92389    0.32110  5.9915 2.079e-09 ***
ma5  0.43838    0.11573  3.7881 0.0001518 ***
```

```
ma3  0.45656    0.11575  3.7661 0.0001516
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

In [537]:

```
tsdiag(arima2)
```



In [538]:

```
Box.test(arima2$residuals,lag=6)
```

```
 Box-Pierce test

data:  arima2$residuals
X-squared = 5.0819, df = 6, p-value = 0.5333
```

In [539]:

```
Box.test(arima2$residuals,lag=12)
```

```
 Box-Pierce test

data:  arima2$residuals
X-squared = 5.5326, df = 12, p-value = 0.9378
```

In [540]:

```
Box.test(arima2$residuals,lag=18)
```

```
 Box-Pierce test

data:  arima2$residuals
X-squared = 6.3501, df = 18, p-value = 0.9946
```

In [541]:

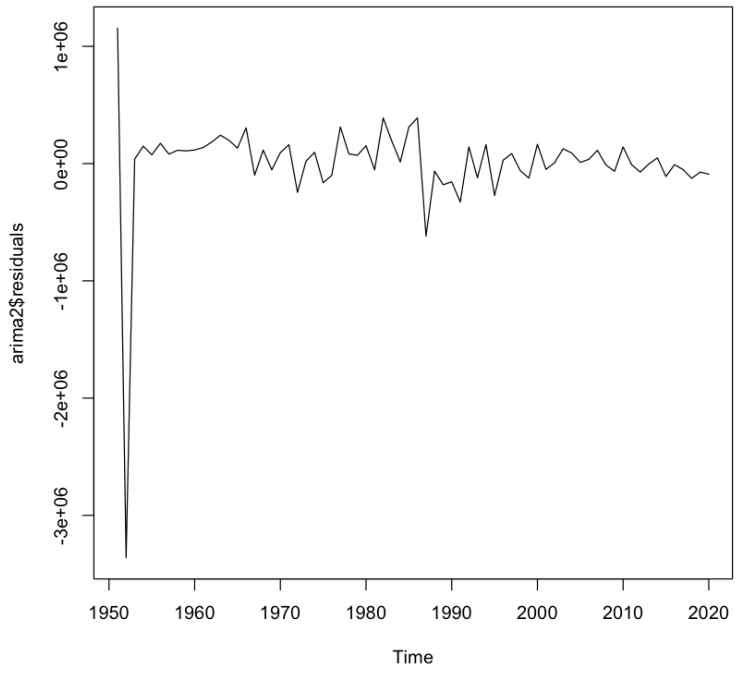
```
Box.test(arima2$residuals,lag=24)
```

```
 Box-Pierce test

data:  arima2$residuals
X-squared = 7.2655, df = 24, p-value = 0.9996
```

In [542]:

```
plot.ts(arima2$residuals)
```



In [543]:

```
arima3=arima(datats,order=c(2,2,5))
arima3
```

```
Call:
arima(x = datats, order = c(2, 2, 5))

Coefficients:
         ar1      ar2     ma1     ma2     ma3     ma4     ma5
      1.1134  -0.4233  1.4580  1.3380  1.4209  1.4264  0.4065
s.e.  0.1598   0.1515  0.1638  0.2297  0.2095  0.2071  0.1585

sigma^2 estimated as 1.847e+10:  log likelihood = -909.27,  aic = 1834.53
```

In [544]:

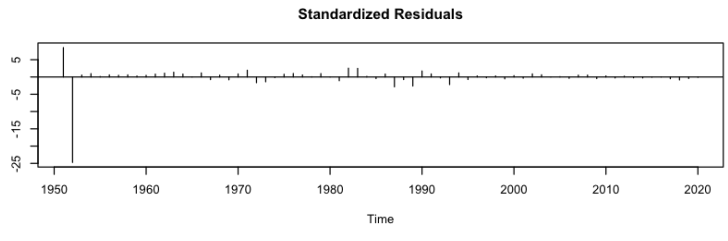
```
coeftest(arima3)
```

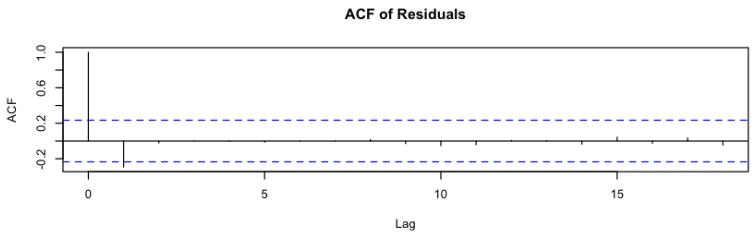
```
z test of coefficients:

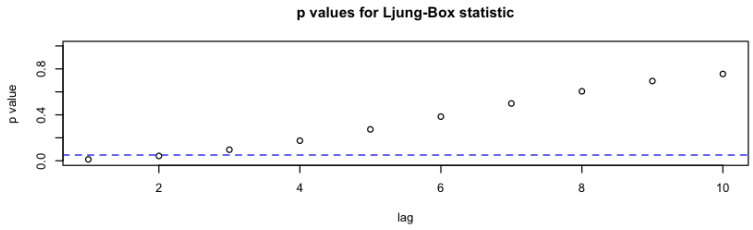
    Estimate Std. Error z value  Pr(>|z|)
ar1  1.11336    0.15984  6.9654 3.274e-12 ***
ar2 -0.42330    0.15154 -2.7933  0.005217 **
ma1  1.45799    0.16380  8.9011 < 2.2e-16 ***
ma2  1.33804    0.22967  5.8260 5.678e-09 ***
ma3  1.42088    0.20946  6.7836 1.172e-11 ***
ma4  1.42643    0.20706  6.8891 5.615e-12 ***
ma5  0.40652    0.15852  2.5644  0.010334 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

In [545]:

```
tsdiag(arima3)
```

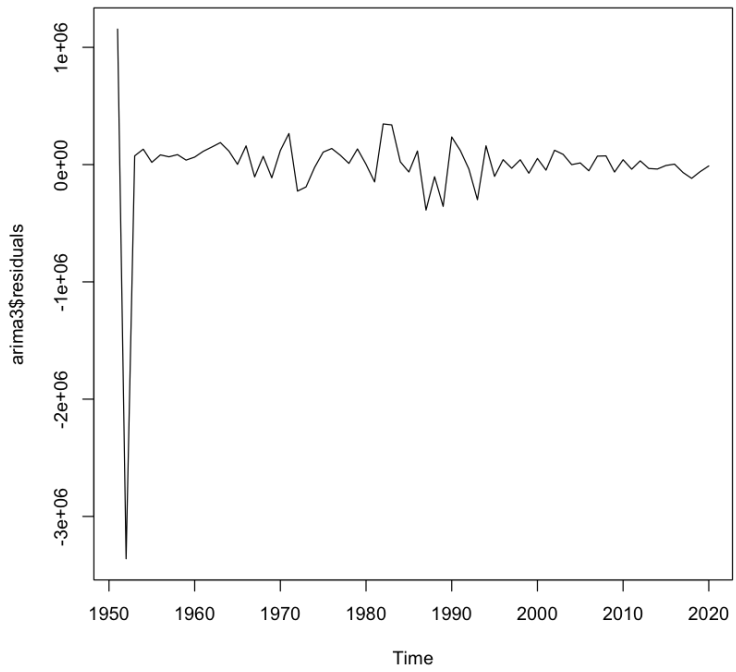






In [546]:

```
plot.ts(arima3$residuals)
```



In [547]:

```
library(forecast)
auto.arima(datats)
```

```
Series: datats
ARIMA(3,2,1)

Coefficients:
         ar1      ar2     ar3     ma1
      1.9813  -1.4340  0.3824  0.3980
s.e.  0.1469   0.2634  0.1406  0.1339

sigma^2 = 2.344e+11:  log likelihood = -924.8
AIC=1859.61   AICc=1860.58   BIC=1870.71
```

In [548]:

```
arima4=arima(datats,order=c(3,2,1))
arima4
```

```
Call:
arima(x = datats, order = c(3, 2, 1))
```

```
Coefficients:
         ar1      ar2     ar3     ma1
      1.9813  -1.4340  0.3824  0.3980
s.e.  0.1469   0.2634  0.1406  0.1339

sigma^2 estimated as 3.477e+10:  log likelihood = -924.8,  aic = 1859.61
```

In [549]:

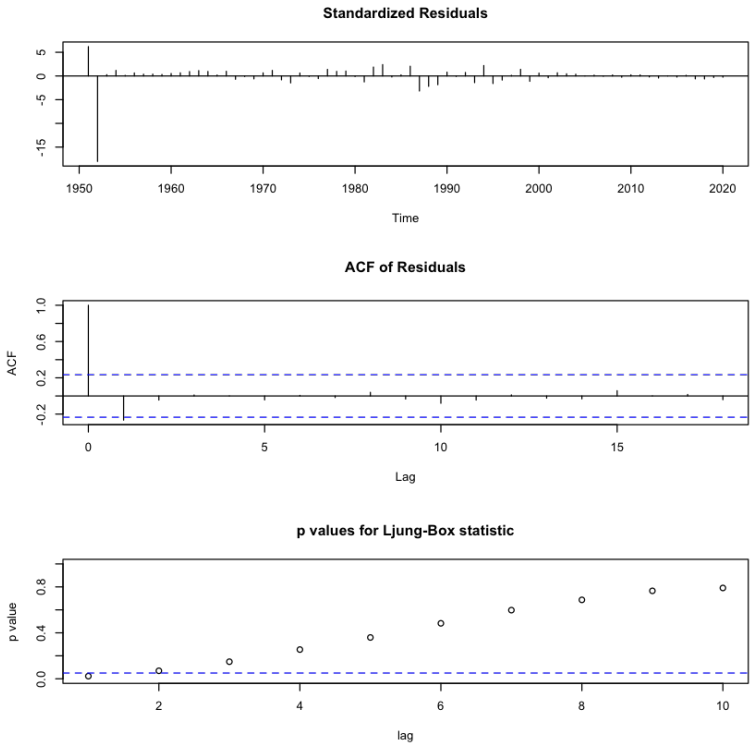
```
coeftest(arima4)
```

```
z test of coefficients:

    Estimate Std. Error z value  Pr(>|z|)
ar1  1.98126    0.14691 13.4864 < 2.2e-16 ***
ar2 -1.43399    0.26336 -5.4451 5.178e-08 ***
ar3  0.38237    0.14064  2.7189  0.006550 **
ma1  0.39796    0.13387  2.9726  0.002953 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

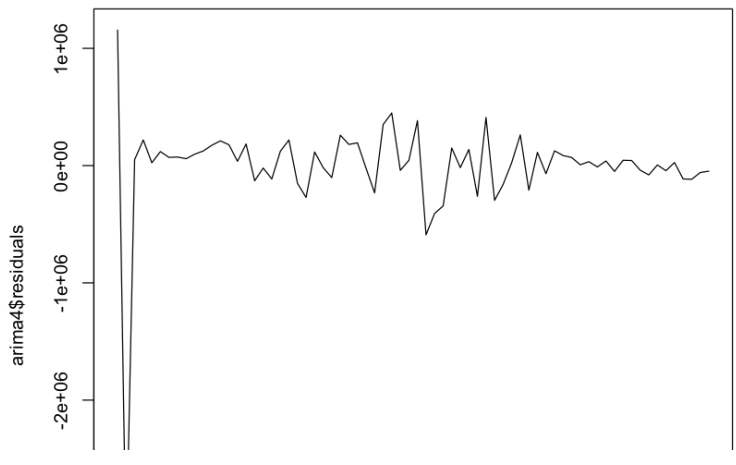
In [550]:

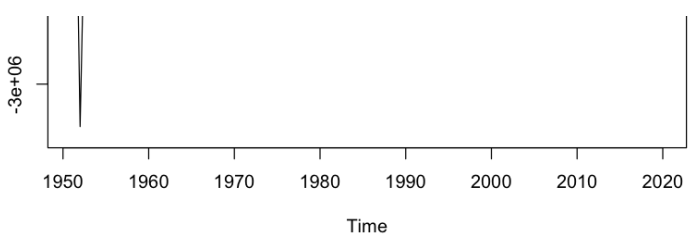
```
tsdiag(arima4)
```



In [551]:

```
plot.ts(arima4$residuals)
```




-3e+06
1950
1960
1970
1980
1990
2000
2010
2020
Time